

Trainings ▾

Preparatory Steps

Automotive with CM870

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>)
- Remove the turbocharger oil supply and oil drain lines from the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html>)

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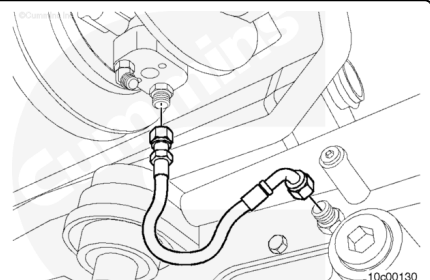
- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>)
- Remove the turbocharger oil drain line from the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html>)

Remove

Automotive with CM870

Use a second wrench to hold the coolant hose coupling while loosening the line.

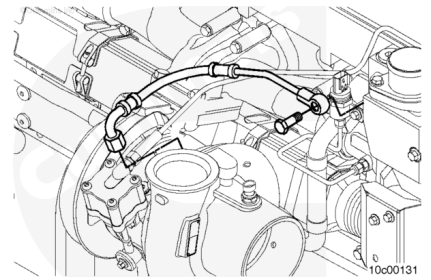
Remove the turbocharger coolant supply line from the bottom of the turbocharger and the lubricating oil filter head.



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Use a second wrench to hold the coolant hose coupling while loosening the line.

Remove the turbocharger coolant return line from the top of the turbocharger and the thermostat housing.

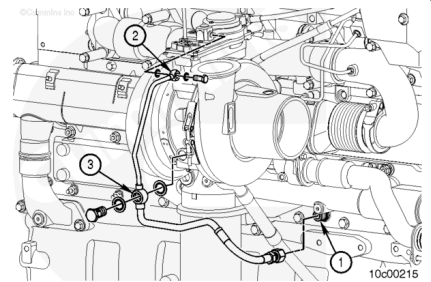


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Use a second wrench to hold the coolant hose coupling while loosening the line.

Remove the turbocharger coolant supply hose from the lubricating oil filter head (1).

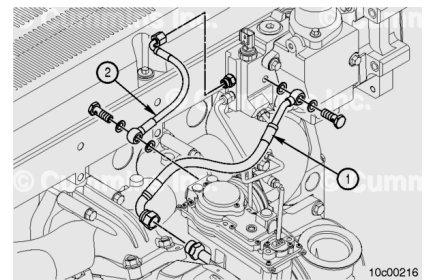
Remove the smaller banjo bolt from the turbocharger electric actuator coolant inlet port (2) and the large banjo bolt from the turbocharger bearing housing coolant inlet port (3).



Use a second wrench to hold the coolant hose coupling while loosening the line.

Remove the turbocharger bearing housing coolant return hose from the top of the turbocharger and the thermostat housing (1).

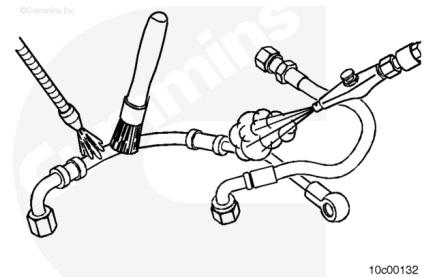
Remove the turbocharger actuator coolant return hose from the top of the turbocharger and the thermostat housing (2).



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

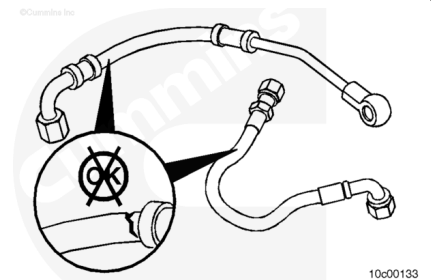
Clean the turbocharger coolant hoses with solvent.

Dry with compressed air.

Check the o-rings for cuts or deformation. Replace as necessary.

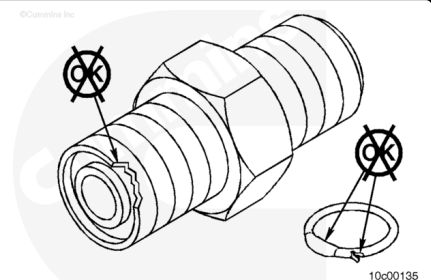
Check the turbocharger coolant hoses for cracks or other damage.

Replace if damage is found.



Check the face of the coolant fittings on the engine and on the turbocharger to be sure the o-ring is in place and is **not** damaged.

A damaged or missing o-ring will result in a coolant leak.



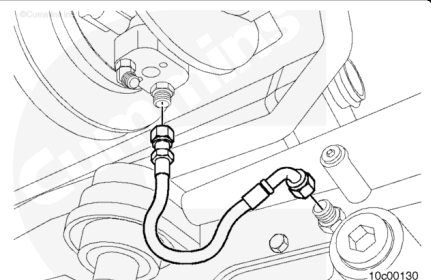
Install

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⚠ CAUTION ⚠

Do not bend or kink hoses while tightening. Doing so can damage the hoses.

Install the turbocharger coolant supply hose to the bottom of the turbocharger and the lubricating oil filter head. Tighten the fittings.

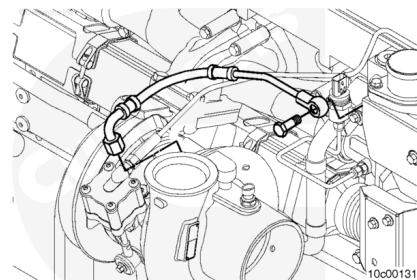


Use a second wrench to hold the coolant hose coupling while tightening the line.

Torque Value: 46 n•m [34 ft-lb]

⚠ CAUTION ⚠

Do not bend or kink hoses while tightening. Doing so can damage the lines.



Check the face of the coolant fittings on the engine and on the turbocharger to be sure the o-ring is in place and is **not** damaged.

A damaged or missing o-ring will result in a coolant leak.

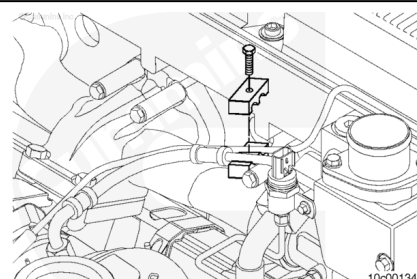
Install the turbocharger coolant return hose to the top of the turbocharger and the thermostat housing. Tighten the fittings.

Use a second wrench to hold the coolant hose coupling while tightening the line.

Torque Value: 46 n•m [34 ft-lb]

Install the tube brace onto the turbocharger coolant return hose and the air tube.

Torque Value: 18 n•m [159 in-lb]



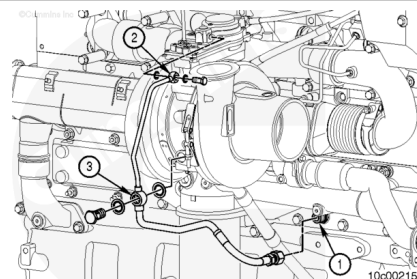
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⚠ CAUTION ⚠

Do not bend or kink hoses while tightening. Doing so can damage the hoses.

⚠ CAUTION ⚠

Do not use petroleum products on the turbocharger coolant hose o-rings. The o-rings will swell and cause the connection to leak.



Lubricate the sealing washers with clean engine coolant, soapy water, or vegetable oil.

Install the turbocharger coolant supply hose to the turbocharger bearing housing (3).

Torque Value: 41 n•m [30 ft-lb]

Install the turbocharger coolant supply hose to the turbocharger actuator (2).

Torque Value: 15 n•m [133 in-lb]

Install the turbocharger coolant supply hose to the lubricating oil filter head (1).

Torque Value: 54 n•m [40 ft-lb]

Note : Use a second wrench to hold the coolant hose coupling while tightening the hose.

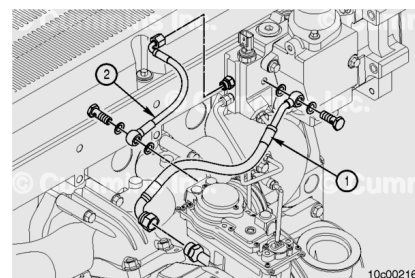
Tighten the fittings.

⚠ CAUTION ⚠

Do not bend or kink hoses while tightening. Doing so can damage the lines.

⚠ CAUTION ⚠

Do not use petroleum products on the turbocharger coolant line o-rings. The o-rings will swell and cause the connection to leak.



Install the turbocharger coolant return hose to the turbocharger. Tighten the fitting.

Torque Value: 46 n•m [34 ft-lb]

Install the turbocharger coolant return hose in the thermostat housing. Tighten the banjo fitting.

Torque Value: 24 n•m [18 ft-lb]

Install the turbocharger actuator coolant hose to the thermostat housing. Tighten the fitting.

Torque Value: 16 n•m [142 in-lb]

It is possible for the turbocharger actuator coolant return hose to contact the turbocharger coolant return hose when the fittings are tightened. Be sure to space the hoses apart while the fittings are tightened. If the hoses are **not** spaced apart, hose failure can result.

Note : Use a second wrench to hold the coolant line coupling while tightening the line.

Install the turbocharger actuator coolant hose to the turbocharger actuator. Tighten the banjo fitting.

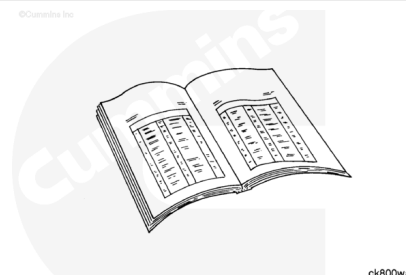
Torque Value: 24 n·m [18 ft-lb]

Finishing Steps

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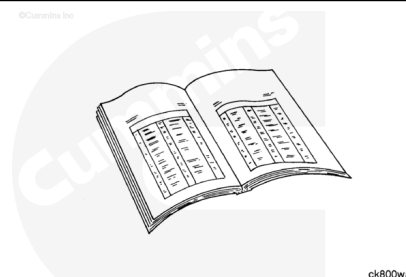


- Install the turbocharger oil supply and oil drain hoses from the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html>)
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/10/10-013-009.html>)
- Start the engine and check for leaks.

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- Install the turbocharger oil drain line from the turbocharger. Refer to Procedure 010-033 (</qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html>).

- Fill the engine with coolant. Refer to Procedure 008-018. (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
- Connect the batteries. Refer to Procedure 013-009 (/qs3/pubsys2/xml/en/procedures/10/10-013-009.html).
- Start the engine and check for leaks.

Last Modified: 08-Jul-2015
